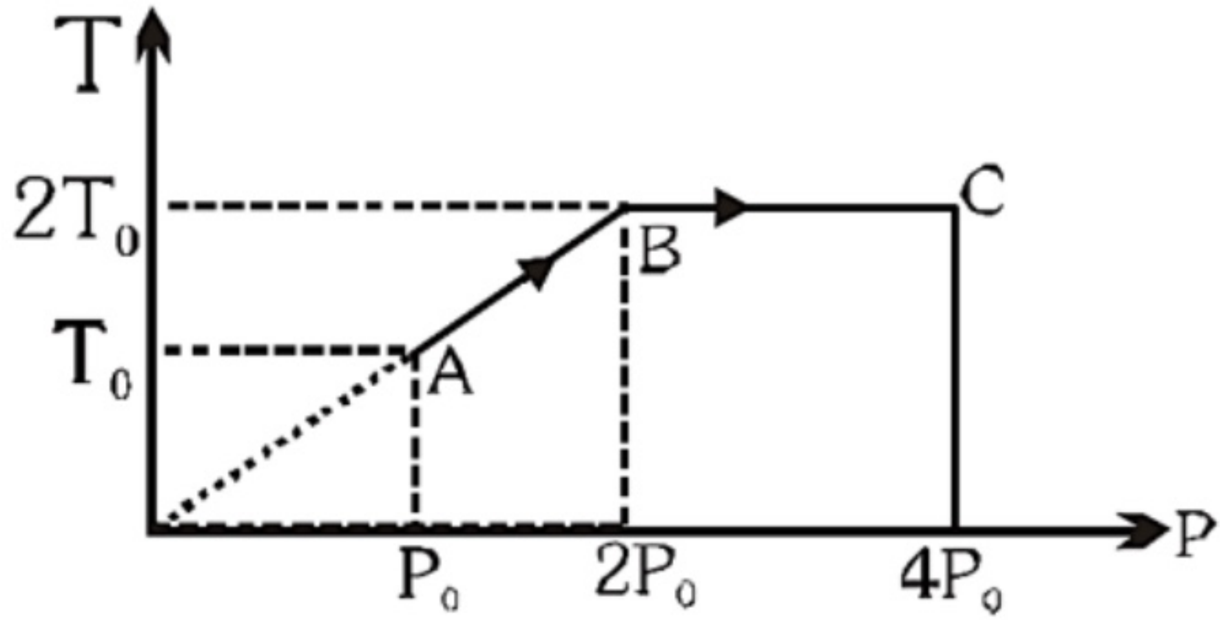
DROPPER-NEET-RAJPUR-ROAD-CENTRE-TEST — with answer key

180 questions · 180 minutes · marking +4 / -1 / 0

Q1. Single correct

One mole of an ideal gas is taken through the process ABC as shown in the figure. The total work done on the gas is:



(A) zero

(B) $2RT_0 \ln 2$ ✓

(C) $-2RT_0 \ln 2$

(D) $4RT_0 \ln 2$

Q2. Single correct

A body of mass 0.1 kg attains a velocity of $10 \mathrm{~ms}^{-1}$ in 0.1 s. The force acting on the body is : [AIPMT screening (Med.) 2006]

(A) 10 N ✓

(B) 0.01 N

(C) 0.1 N

(D) 100 N

Q3. Single correct

A body, under the action of a force $\vec{F}=6 \hat{i}-8 \hat{j}+10 \hat{k}$, acquires an acceleration of $1 \mathrm{~ms}^{-2}$. The mass of this body must be [AIPMT screening (Med.) 2009]

(A) $2 \sqrt{10}$ kg ✓

(B) 10 kg

(C) 20 kg

(D) $10 \sqrt{2}$ kg

Q4. Single correct

2 moles of an ideal monoatomic gas occupying volume V is adiabatically expanded from temperature 300K to a volume of $2 \sqrt{2} V$. Then the final temperature & change in internal energy are respectively $(R=8.3)$

(A) 150 K, -3735 J ✓

(B) 140 K, -3735 J

(C) 150 K, -3537 J

(D) 140 K, -3537 J

Q5. Single correct

The molar heat capacity of a monoatomic ideal gas undergoing the process $PV^{1/2} = \text{constant}$ is

(A) $\frac{15}{2}R$

(B) $\frac{23}{2}R$

(C) $\frac{7}{2}R$ ✓

(D) Zero

Q6. Single correct

A gas is found to obey $P^2V = \text{constant}$. The initial temperature and volume of gas are T_0 & V_0. If the gas expands to volume $3V_0$. Find the final temperature of gas.

(A) T_0

(B) $\sqrt{3}T_0$ ✓

(C) $2T_0$

(D) $\frac{T_0}{2}$

Q7. Single correct

A person of mass 60 kg is inside a lift of mass 940 kg and presses the button on control panel. The lift starts moving upwards with an acceleration $1.0\,\mathrm{m/s^2}$. If $g = 10\,\mathrm{ms^{-2}}$, the tension in the supporting cable is : [AIPMT Screening-2011]

(A) 8600 N

(B) 9680 N

(C) 11000 N ✓

(D) 1200 N

Q8. Single correct

A mixture of ideal gasses N_2 and He are taken in the mass ratio of 14 : 1 respectively. Molar heat capacity of the mixture at constant pressure is :-

(A) $\frac{19R}{6}$ ✓

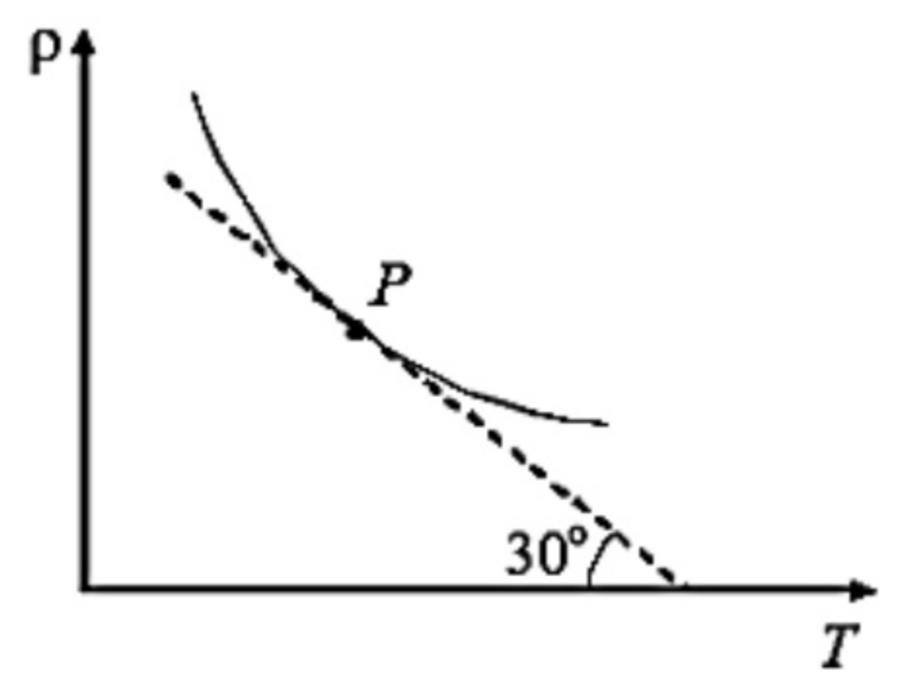
(B) $\frac{6R}{19}$

(C) $\frac{13R}{6}$

(D) $\frac{6R}{13}$

Q9. Single correct

The variation of density (ρ) with temperature (T) of an ideal gas of molecular mass (M) at constant pressure is plotted as shown in figure. The pressure of the gas at state P will be :-



(A) $\frac{RT^2}{\sqrt{3}M}$ ✓

(B) $\frac{\sqrt{3}RT^2}{M}$

(C) $\frac{RT^2}{M}$

(D) $\frac{RT^2}{2M}$

Q10. Single correct

Temperature of mixture of 2 moles of an ideal diatomic gas and 3 moles of an ideal monoatomic gas is increased by $3^{\circ}\mathrm{C}$ at constant pressure. What is the work done by the gas mixture? (R is universal gas constant)

(A) 18 R

(B) 24 R

(C) 15 R ✓

(D) 27 R

Q11. Single correct

A spring of force constant k is cut into lengths of ratio 1 : 2 : 3. They are connected in series and the new force constnt is k'. Then they are connected in parallel and force constant is k''. Then k' : k'' is : [NEET2017]

(A) 1 : 6

(B) 1 : 9

(C) 1 : 11 ✓

(D) 1 : 14

Q12. Single correct

When an object is shot from the bottom of a long smooth inclined plane kept at an angle 60° with horizontal, it can travel a distance x_1 along the plane. But when the inclination is decreased to 30° and the same object the shot with the same velocity, it can travel x_2 distance. Then $x_1 : x_2$ will be [NEET (UG) 2019]

(A) $1 : \sqrt{2}$

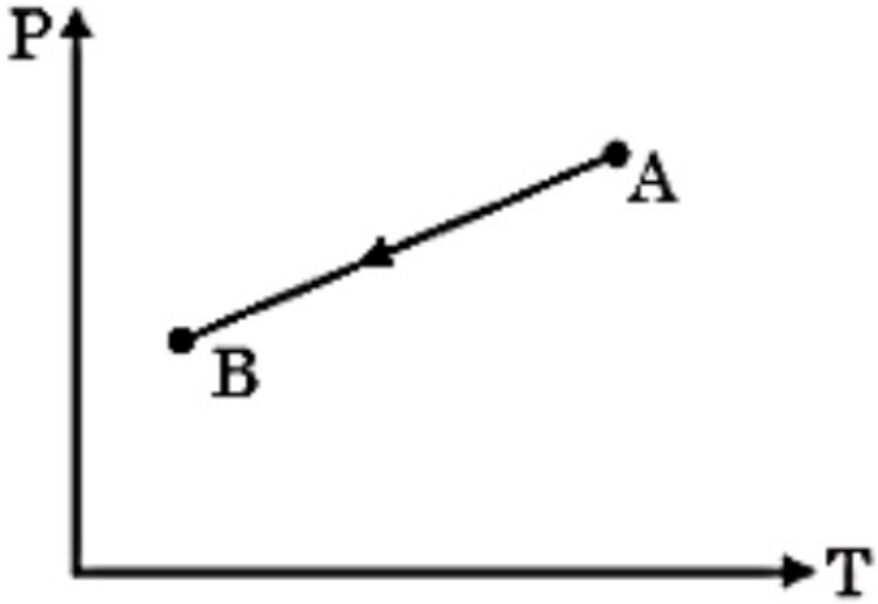
(B) $\sqrt{2} : 1$

(C) $1 : \sqrt{3}$ ✓

(D) $1 : 2\sqrt{3}$

Q13. Single correct

An ideal gas undergoes a process $A \rightarrow B$ as shown on the pressure-temperature graph. The gas undergoes :-



(A) Compression ✓

(B) Expansion

(C) No change in volume

(D) Data insufficient

Q14. Single correct

A force acting on a particle moving in the xy-plane is given by $\vec{F} = (2y\hat{i} + x^2\hat{j})\,N$, where x and y are in m. The particle moves along a straight line from the origin to (5, 5). The work done by F is :

(A) 125 J

(B) 66.7 J ✓

(C) 35 J

(D) 25 J

Q15. Single correct

Select the INCORRECT statement :-

(A) Work done by static friction on an object may be positive.

(B) Work done by a force depends on frame of reference.

(C) Kinetic friction may perform positive work on an object.

(D) Work done by gravitational force depend on path followed by object. ✓

Q16. Single correct

A body of mass 10 kg is released from a tower of height 20m and body acquires a velocity of $10 \mathrm{ms}^{-1}$ after falling through the distance 20m. The work done by the push of the air on the body is:- (Take $g = 10\,\mathrm{m/s}^2$)

(A) 1500 J

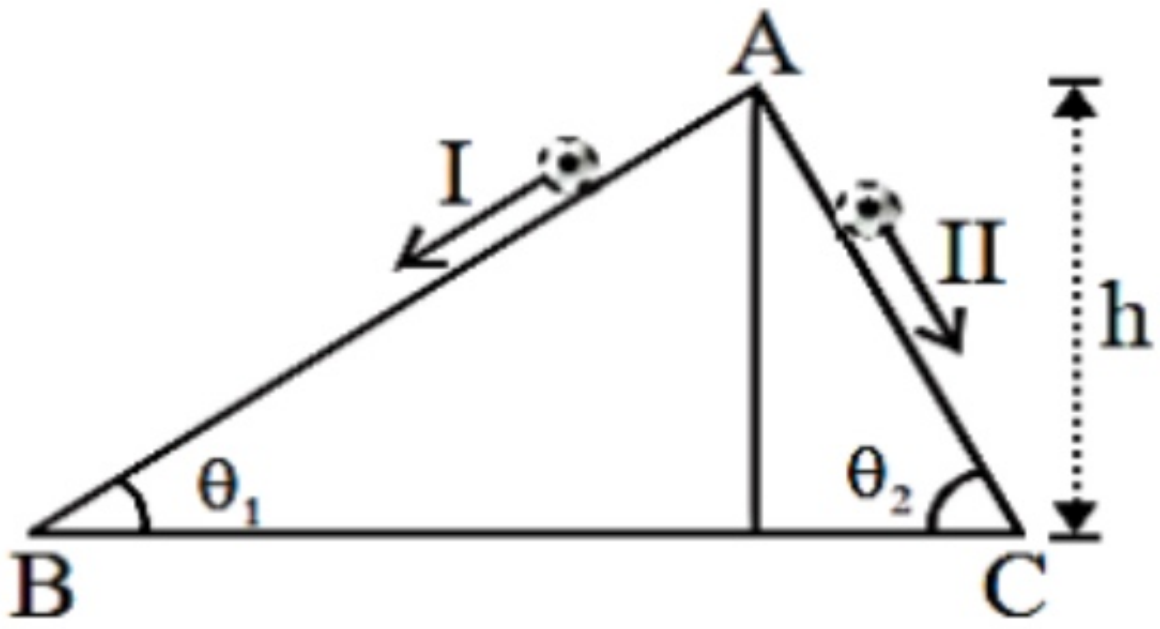
(B) 1800 J

(C) -1500 J ✓

(D) -1800 J

Q17. Single correct

Two inclined frictionless tracks, one gradual and the other steep meet at A from where two stones are allowed to slide down from rest, one on each track as shown in figure. Which of the following statement is correct?



(A) Both the stones reach the bottom at the same time but not with the same speed.

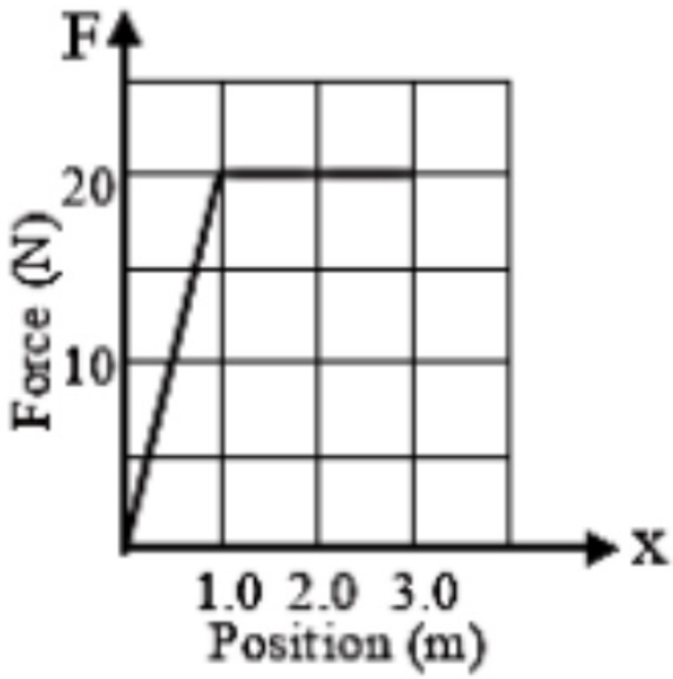
(B) Both the stones reach the bottom with the same speed and stone I reaches the bottom earlier than stone II.

(C) Both the stones reach the bottom with the same speed and stone II reaches the bottom earlier than stone I. ✓

(D) Both the stones reach the bottom at different times and with different speeds.

Q18. Single correct

The graph below shows how the force on a mass depends on the position of the mass. What is the change in the kinetic energy of the mass as it moves from $x = 0.0\,m$ to $x = 3.0\,m$?



(A) 0.0 J

(B) 20 J

(C) 50 J ✓

(D) 60 J

Q19. Single correct

Potential energy of a particle is $U = \frac{x^3}{3} - \frac{5x^2}{2} + 4x - 3$. If it is not experiencing any other force, then find position where it is in stable equilibrium.

(A) $x = 1$

(B) $x = -1$

(C) $x = 4$ ✓

(D) none

Q20. Single correct

A toy car can deliver a constant power of 20W. The resistive force on the car is αv, where v is velocity of car in m/s. If maximum velocity of car is 2m/s, the value of α is.

(A) 10

(B) 5 ✓

(C) 15

(D) None of these

Q21. Single correct

A man is standing in the middle of a perfectly smooth 'island of ice' where there is no friction between the ground and his feet. Under these circumstances

(A) he can reach the desired corner by throwing any object in the same direction

(B) he can reach the desired corner by throwing any object in the opposite direction ✓

(C) he has no chance of reaching any corner of the island

(D) he can reach the desired corner by pursuing on the ground in that direction

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Q22. Single correct

A block of mass 20 kg is pushed with a horizontal force of 90N. If the coefficient of static and kinetic friction are 0.4 and 0.3, the frictional force acting on the block is $(g = 10\mathrm{ms}^{-2})$

(A) 90N

(B) 80N

(C) 60N ✓

(D) 30N

Q23. Single correct

Power applied to a particle varies with time as $P = (3t^2 - 2t + 1)W$, where t is in second. Find the change in its kinetic energy between time $t = 2s$ and $t = 4s$.

(A) 32 J

(B) 46 J ✓

(C) 61 J

(D) 102 J

Q24. Single correct

A body of mass m is accelerated uniformly from rest to a speed v in a time T. The instantaneous power delivered to the body as a function of time, is given by

(A) $\frac{mv^2}{T^2}t$ ✓

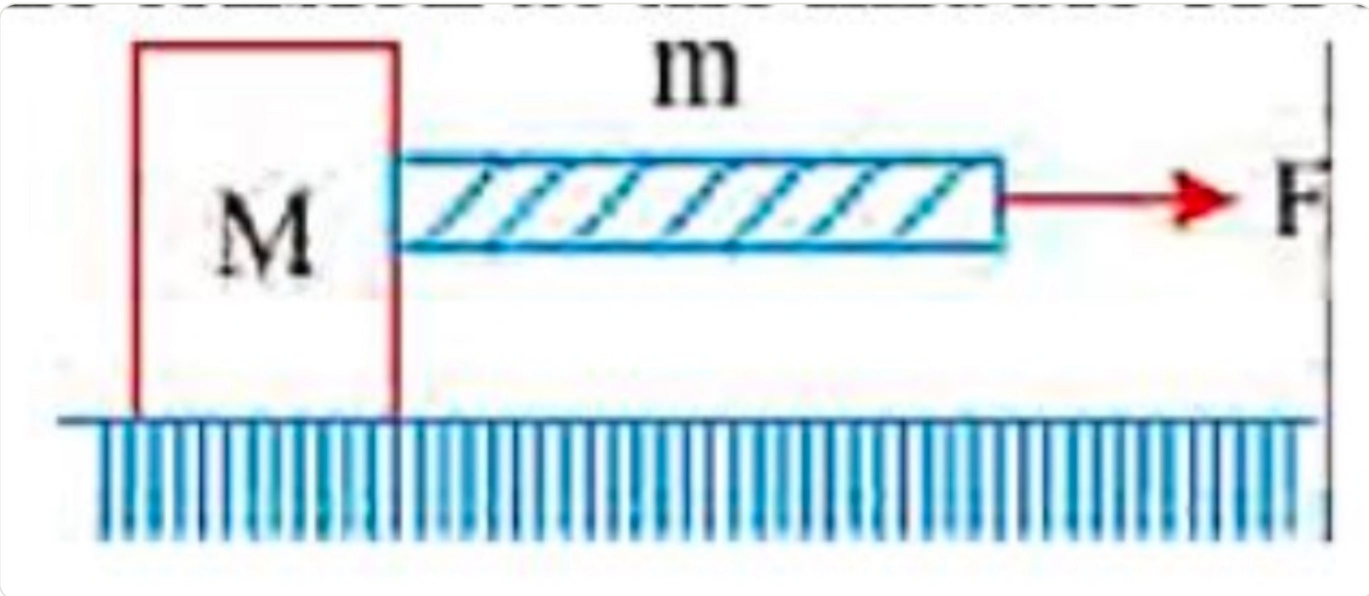
(B) $\frac{mv^2}{T^2}t^2$

(C) $\frac{1}{2}\frac{mv^2}{T^2}t$

(D) $\frac{1}{2}\frac{mv^2}{T^2}t^2$

Q25. Single correct

The block is placed on a frictionless surface in gravity free space. A heavy string of a mass m is connected and force F is applied on the string, then the tension at the middle of rope is



(A) $\frac{\left(\frac{m}{2}+M\right)F}{m+M}$ ✓

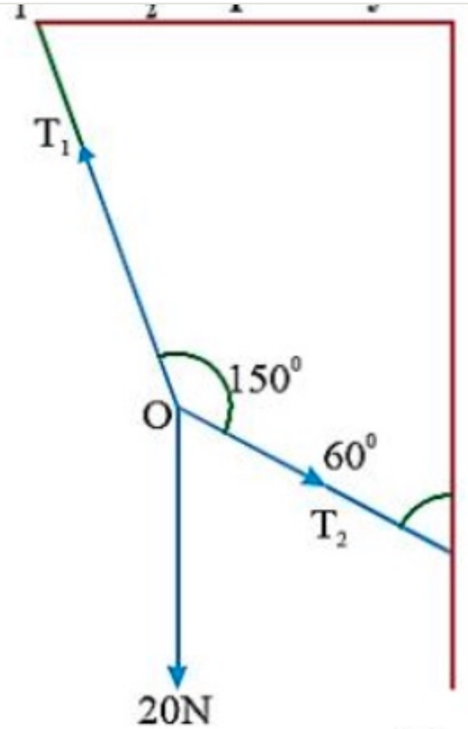
(B) $\frac{\left(\frac{M}{2}+m\right)F}{m+M}$

(C) zero

(D) $\frac{MF}{m+M}$

Q26. Single correct

If 'O' is at equilibrium then the values of the tension T_1 and T_2 respectively.



(A) 20N, 30N

(B) $20\sqrt{3}$N, 20N ✓

(C) $20\sqrt{3}$N, $20\sqrt{3}$N

(D) 10N, 30N

Q27. Single correct

The minimum force required to start pushing a body up a rough (frictional coefficient μ) inclined plane is F_1 while the minimum force needed to prevent it from sliding down is F_2. If the inclined plane makes an angle θ with the horizontal such that $\tan\theta = 2\mu$, then the ratio $\frac{F_1}{F_2}$ is

(A) 4

(B) 1

(C) 2

(D) 3 ✓

Q28. Single correct

While walking on ice, one should take small steps to avoid slipping. This is because smaller steps ensure

(A) larger friction

(B) smaller friction ✓

(C) larger normal force

(D) smaller normal force.

Q29. Single correct

Two cars of unequal masses use similar tyres. If they are moving at the same initial speed, the minimum stopping distance

(A) is smaller for the heavier car

(B) is smaller for the lighter car

(C) is same for both cars ✓

(D) depends on the volume of the car.

Q30. Single correct

A car accelerates on a horizontal road due to the force exerted by

(A) the engine of the car

(B) the driver of the car

(C) the earth

(D) the road. ✓

Q31. Single correct

The molar specific heat at constant pressure of an ideal gas is $(7/2)R$. The ratio of specific heat at constant pressure to that at constant volume is : [AIPMT-2006]

(A) 7/5 ✓

(B) 8/7

(C) 5/7

(D) 9/7

Q32. Single correct

The translational kinetic energy of molecules of one mole of a monoatomic gas is $U = 3NKT/2$. The value of atomic specific heat of gas under constant pressure will be : [AIPMT 2006]

(A) $\frac{3}{2}R$

(B) $\frac{5}{2}R$ ✓

(C) $\frac{7}{2}R$

(D) $\frac{9}{2}R$

Q33. Single correct

If Q, E and W denote respectively the heat added, change in internal energy and the work done in a closed cycle process, then :- [AIPMT 2008]

(A) E = 0 ✓

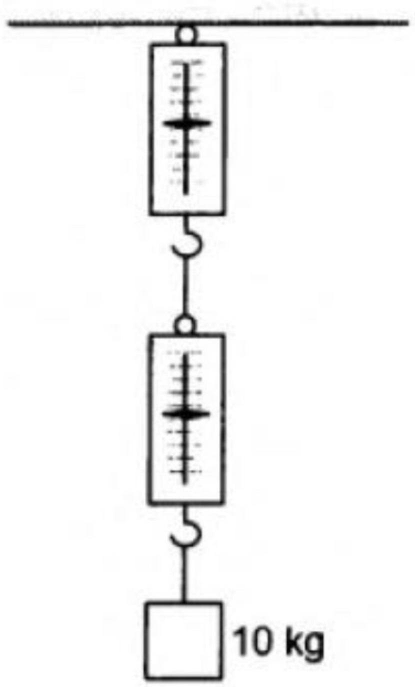
(B) Q = 0

(C) W=0

(D) Q=W=0

Q34. Single correct

A block of mass 10 kg is suspended through two light spring balances as shown in figure (5-Q2).



(A) Both the scales will read 10 kg. ✓

(B) Both the scales will read 5 kg.

(C) The upper scale will read 10 kg and the lower zero.

(D) The readings may be anything but their sum will be 10 kg.

Q35. Single correct

In thermodynamic processes which of the following statement is not true : [AIPMT 2009]

(A) In an adiabatic process $PV^{\gamma} = \text{constant}$

(B) In an adiabatic process the system is insulated from the surroundings

(C) In an isochoric process pressure remains constant ✓

(D) In an Isothermal process the temperature remains constant

Q36. Single correct

The mean square speed of the molecules of a gas at absolute temperature T is proportional to

(A) $\frac{1}{T}$

(B) $\sqrt{T}$

(C) T ✓

(D) T^2

Q37. Single correct

If ΔU and ΔW represent the increase in internal energy and work done by the system respectively in a thermodynamical process, which of the following is true ? [AIPMT 2010]

(A) $\Delta U = -\Delta W$, in a isothermal process

(B) $\Delta U = -\Delta W$, in a adiabatic process ✓

(C) $\Delta U = \Delta W$, in a isothermal process

(D) $\Delta U = \Delta W$, in a adiabatic process

Q38. Single correct

A monoatomic gas at pressure P_1 and volume V_1 is compressed adiabatically to 1/8th its original volume. What is the final pressure of the gas : [AIPMT 2010]

(A) P_1

(B) $16P_1$

(C) $32P_1$ ✓

(D) $64P_1$

Q39. Single correct

Suppose a container is evacuated to leave just one molecule of a gas in it. Let v_a and v_{rms} represent the average speed and the rms speed of the gas.

(A) $v_a > v_{rms}$

(B) $v_a < v_{rms}$

(C) $v_a = v_{rms}$ ✓

(D) v_{rms} is undefined.

Q40. Single correct

During an isothermal expansion, a confined ideal gas does +150 J of work in its surroundings. This implies that :- [AIPMT 2011]

(A) 150 J of heat has been removed from the gas

(B) 300 J of heat has been added to the gas

(C) No heat is transferred because the process is isothermal

(D) 150 J of heat has been added to the gas ✓

Q41. Single correct

A mass of diatomic gas $(\gamma = 1.4)$ at a pressure of 2 atmospheres is compressed adiabatically so that its temperature rises from $27^\circ\mathrm{C}$ to $927^\circ\mathrm{C}$. The pressure of the gas in the final state is: [AIPMT 2011]

(A) 8 atm

(B) 28 atm

(C) 68.7 atm

(D) 256 atm ✓

Q42. Single correct

The rms speed of oxygen molecules in a gas is v. If the temperature is doubled and the oxygen molecules dissociate into oxygen atoms, the rms speed will become

(A) v

(B) $v\sqrt{2}$

(C) $2v$ ✓

(D) $4v$

Q43. Single correct

During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its temperature. The ratio of C_p/C_v for the gas is : [NEET 2013]

(A) 2

(B) 5/3

(C) 3/2 ✓

(D) 4/3

Q44. Single correct

The mean free path of molecules of gas (radius 'r') is inversely proportional to [AIPMT-2014]

(A) r

(B) $\sqrt{r}$

(C) r^3

(D) r^2 ✓

Q45. Single correct

The ratio of the specific heats $C_p/C_v = \gamma$ in terms of degrees of freedom (n) is given by : [AIPMT NEET 2015]

(A) $\left(1+\frac{n}{3}\right)$

(B) $\left(1+\frac{2}{n}\right)$ ✓

(C) $\left(1+\frac{n}{2}\right)$

(D) $\left(1+\frac{1}{n}\right)$

Q46. Single correct

The measurement of the electron position is associated with an uncertainty in momentum, which is equal to $1 \times 10^{-18}\ \mathrm{g\ cm\ s^{-1}}$, the uncertainty in electron velocity is:
(mass of electron = $9 \times 10^{-28}\mathrm{g}$)

(A) $1 \times 10^{11}\ \mathrm{cm\ s^{-1}}$

(B) $1 \times 10^{9}\ \mathrm{cm\ s^{-1}}$ ✓

(C) $1 \times 10^{6}\ \mathrm{cm\ s^{-1}}$

(D) $1 \times 10^{5}\ \mathrm{cm\ s^{-1}}$

Q47. Single correct

Which of the following is not possible arrangement of electrons in an atom?

(A) $n=3,\ \ell=2,\ m=-2,\ s=-1/2$

(B) $n=4,\ \ell=0,\ m=0,\ s=-1/2$

(C) $n=5,\ \ell=3,\ m=0,\ s=+1/2$

(D) $n=3,\ \ell=2,\ m=-3,\ s=-1/2$ ✓

Q48. Single correct

The energies E_1 and E_2 of two radiations are 25 eV and 50 eV respectively. The relation between their wavelengths i.e. λ_1 and λ_2 will be:

(A) $\lambda_1 = \lambda_2$

(B) $\lambda_1 = 2\lambda_2$ ✓

(C) $\lambda_1 = 4\lambda_2$

(D) $\lambda_1 = \lambda_2$

Q49. Single correct

Maximum number of electrons in a subshell with $\ell = 3$ and $n = 4$ is:

(A) 10

(B) 12

(C) 14 ✓

(D) 16

Q50. Single correct

The orbital angular momentum of a p-electron is given as:

(A) $\sqrt{\frac{2h}{2\pi}}$

(B) $\sqrt{6}\,\frac{h}{2\pi}$

(C) $\frac{h}{\sqrt{2}\pi}$ ✓

(D) $\sqrt{3}\,\frac{h}{2\pi}$

Q51. Single correct

Based on equation $E = -2.178 \times 10^{-18}$ J certain conclusions are written. Which of them is not correct?

(A) For n = 1, the electron has a more negative energy than it does for n = 6 which means that the electron is more loosely bound in the smallest allowed orbit. ✓

(B) The negative sign in equation simply means that the energy of electron bound to the nucleus is lower than it would be if the electrons were at the infinite distance from the nucleus

(C) Larger the value of n, the larger is the orbit radius

(D) Equation can be used to calculate the change in energy when the electron change orbit

Q52. Single correct

What is the maximum number of orbitals that can be identified with the following quantum numbers?

$n = 3,\ \ell = 1,\ m = 0$

(A) 1 ✓

(B) 2

(C) 3

(D) 4

Q53. Single correct

Magnetic moment 2.84 B.M. is given by:
(At. no.], Ni = 28, Ti = 22, Cr = 24, Co = 27)

(A) Ti^{3+}

(B) Cr^{2+}

(C) Co^{2+}

(D) Ni^{2+} ✓

Q54. Single correct

The number of d-electrons in Fe^{2+} (Z = 26) is not equal to the number of electrons in which one of the following?

(A) p-electrons in Cl (Z = 17) ✓

(B) d-electrons in Fe (Z = 26)

(C) p-electrons in Ne (Z = 10)

(D) s-electrons in Mg (Z = 12)

Q55. Single correct

Two electrons occupying the same orbital are distinguished by

(A) Principal quantum number

(B) Magnetic quantum number

(C) Azimuthal quantum number

(D) Spin quantum number ✓

Q56. Single correct

How many electrons can fit in an orbital for which $n = 3$ and $l = 1$?

(A) 10

(B) 14

(C) 2 ✓

(D) 6

Q57. Single correct

Which one is the wrong statement?

(A) The uncertainty principle is $\Delta E \times \Delta t \geq h/4\pi$

(B) Half-filled and fully filled orbitals have greater stability due to greater exchange energy, greater symmetry and more balanced arrangement.

(C) The energy of 2s orbital is less than the energy of 2p orbital in case of Hydrogen like atoms ✓

(D) de-Broglie's wavelength is given by, where m = mass of the particle, v = velocity of the particle

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Q58. Single correct

Which of the following series of transitions in the spectrum of hydrogen atom falls in visible region?

(A) Lyman series

(B) Balmer series ✓

(C) Paschen series

(D) Brackett series

Q59. Single correct

Orbital having 3 angular nodes and 3 total nodes is:

(A) 5 p

(B) 3 d

(C) 4 f ✓

(D) 6 d

Q60. Single correct

In hydrogen atom, the de Broglie wavelength of an electron in the second Bohr orbit is:
[Given that Bohr radius, $a_0 = 52.9 \text{ pm}$]

(A) 211.6 pm

(B) $211.6\pi \text{ pm}$ ✓

(C) $52.9\pi \text{ pm}$

(D) 105.8 pm

Q61. Single correct

A 0.0020 m aqueous solution of an ionic compound $[\mathrm{Co(NH_3)_5(NO*2)}]\mathrm{Cl}$ freezes at $-0.00732°\mathrm{C}$. Number of moles of ions which 1mol of ionic compound produces on being dissolved in water will be ($K_f = 1.86°\mathrm{C\ m^{-1}}$):

(A) 1

(B) 2 ✓

(C) 3

(D) 4

Q62. Single correct

A solution of sucrose (molar mass = 342 g mol^{-1}) has been prepared by dissolving 68.5 g of sucrose in 1000 g of water. The freezing point of the solution obtained will be:
(K_f for water = $1.86 \text{ K kg mol}^{-1}$)

(A) $-0.570°\text{C}$

(B) $-0.372°\text{C}$ ✓

(C) $-0.520°\text{C}$

(D) $+0.372°\text{C}$

Q63. Single correct

The Van't Hoff factor i for a compound which undergoes dissociation in one solvent and association in other solvent is respectively:

(A) Less than one and greater than one

(B) Less than one and less than one

(C) Greater than one and less than one ✓

(D) Greater than one and greater than one

Q64. Single correct

p_A and p_B are the vapour pressure of pure liquid components, A and B, respectively of an ideal binary solution. If x_A represents the mole fraction of component A, the total pressure of the solution will be.

(A) $p_B + x_A(p_B - p_A)$

(B) $p_B + x_A(p_A - p_B)$ ✓

(C) $p_A + x_A(p_B - p_A)$

(D) $p_A + x_A(p_A - p_B)$

Q65. Single correct

Vapour pressure of chloroform ($\mathrm{CHCl} * 3$) and dichloromethane ($\mathrm{CH} * 2\mathrm{Cl} * 2$) at $25^{\circ}\mathrm{C}$ are 200 mmHg and 415 mmHg respectively. Vapour pressure of the solution obtained by mixing 25.5 g of $\mathrm{CHCl} * 3$ and 40g of $\mathrm{CH} * 2\mathrm{Cl} * 2$ at the same temperature will be: [Molecular mass of $\mathrm{CHCl} * 3$ = 119.5 u and molecular mass of $\mathrm{CH} * 2\mathrm{Cl} * 2$ = 85 u]

(A) 347.9 mmHg ✓

(B) 280.5 mmHg

(C) 173.9 mmHg

(D) 615 mmHg

Q66. Single correct

Of the following 0.10m aqueous solutions, which one will exhibit the largest freezing point depression?

(A) KCl

(B) $\mathrm{C} * 6\mathrm{H} * 12\mathrm{O} * 6$

(C) $\mathrm{Al} * 2(\mathrm{SO} * 4) * 3$ ✓

(D) $\mathrm{K} * 2\mathrm{SO} * 4$

Q67. Single correct

Which one is not equal to zero for an ideal solution:

(A) $\Delta S * \mathrm{mix}$ ✓

(B) $\Delta V * \mathrm{mix}$

(C) $\Delta P = P * \mathrm{observed} - P * \mathrm{raoult}$

(D) $\Delta H * \mathrm{mix}$

Q68. Single correct

What is the mole fraction of the solute in a 1.00 m aqueous solution?

(A) 0.0354

(B) 0.0177 ✓

(C) 0.177

(D) 1.770

Q69. Single correct

Which of the following statement about the composition of the vapour over an ideal a 1 : 1 molar mixture of benzene and toluene is correct? Assume that the temperature is constant at (25°C). {Given: Vapour Pressure Data at 25°C, benzene = 12.8 kPa, Toluene = 3.85 kPa]

(A) The vapour will contain a higher percentage of benzene ✓

(B) The vapour will contain a higher percentage of toluene

(C) The vapour will contain equal amounts of benezene and toluene

(D) Not enough information is given to make a predication

Q70. Single correct

Consider the following liquid - vapour equilibrium.\nLiquid $\rightleftharpoons$ Vapour\nWhich of the following relations is correct?

(A) $\frac{d\ln G}{dT^2} = \frac{\Delta H_v}{RT^2}$

(B) $\frac{d\ln P}{dT} = \frac{-\Delta H_v}{RT}$

(C) $\frac{d\ln P}{dT^2} = \frac{-\Delta H_v}{T^2}$

(D) $\frac{d\ln P}{dT} = \frac{\Delta H_v}{RT^2}$ ✓

Q71. Single correct

Which one of the following is incorrect for ideal solution?

(A) $\Delta P = P * \text{obs} - P * \text{calculated by Raoult's law} = 0$

(B) $\Delta G * \text{mix} = 0$ ✓

(C) $\Delta H * \text{mix} = 0$

(D) $\Delta U * \text{mix} = 0$

Q72. Single correct

If molality of a dilute solution is doubled, the value of molal depression constant (K_f) will be:

(A) halved

(B) tripled

(C) unchanged ✓

(D) doubled

Q73. Single correct

For an ideal solution, the correct option is:

(A) $\Delta * \text{mix} S = 0$ at constant T and P

(B) $\Delta * \text{mix} V \neq 0$ at constant T and P

(C) $\Delta * \text{mix} H = 0$ at constant T and P ✓

(D) $\Delta * \text{mix} G = 0$ at constant T and P

Q74. Single correct

Which of the following statements is correct regarding a solution of two compounds A and B exhibiting positive deviation from ideal behaviour?

(A) Intermolecular attractive forces between A-A and B-B are stronger than those between A-B. ✓

(B) $\Delta * \text{mix} H = 0$ at constant T and P

(C) $\Delta * \text{mix} V = 0$ at constant T and P

(D) Intermolecular attractive forces between A-A and B-B are equal to those between A-B.

Q75. Single correct

The density of 2 M aqueous solution of NaOH is 1.28 g/cm³. The molality of the solution is [Given that molecular mass of $\mathrm{NaOH} = 40 \text{ g mol}^{-1}$]

(A) 1.20 m

(B) 1.56 m

(C) 1.67 m ✓

(D) 1.32 m

Q76. Single correct

For the reaction A + B → products, it is observed that:
(a) on doubling the initial concentration of A only, the rate of reaction is also doubled and
(b) on doubling the initial concentrations of both A and B, there is a change by a factor of 8 in the rate of the reaction.\nThe rate of this reaction is given by:

(A) rate = $k[A][B]$

(B) rate = $k[A]^2[B]$

(C) rate = $k[A][B]^2$ ✓

(D) rate = $k[A]^2[B]^2$

Q77. Single correct

Half-life period of a first-order reaction is 1386 s. The specific rate constant of the reaction is:

(A) $5.0 \times 10^{-2}\ \mathrm{s}^{-1}$

(B) $5.0 \times 10^{-3}\ \mathrm{s}^{-1}$

(C) $0.5 \times 10^{-2}\ \mathrm{s}^{-1}$

(D) $0.5 \times 10^{-3}\ \mathrm{s}^{-1}$ ✓

Q78. Single correct

For the reaction $\mathrm{N_2O_5}(g) \rightarrow 2\mathrm{NO_2}(g) + \mathrm{O_2}(g)$ the value of rate of disappearance of $\mathrm{N_2O_5}$ is given as $6.25 \times 10^{-3}\ \mathrm{mol\ L^{-1}s^{-1}}$. The rate of formation of $\mathrm{NO_2}$ and $\mathrm{O_2}$ is given respectively as:

(A) $1.25 \times 10^{-2}\ \mathrm{mol\ L^{-1}s^{-1}}$ and $6.25 \times 10^{-3}\ \mathrm{mol\ L^{-1}s^{-1}}$

(B) $6.25 \times 10^{-3}\ \mathrm{mol\ L^{-1}s^{-1}}$ and $6.25 \times 10^{-3}\ \mathrm{mol\ L^{-1}s^{-1}}$

(C) $1.25 \times 10^{-2}\ \mathrm{mol\ L^{-1}s^{-1}}$ and $3.125 \times 10^{-3}\ \mathrm{mol\ L^{-1}s^{-1}}$ ✓

(D) $6.25 \times 10^{-3}\ \mathrm{mol\ L^{-1}s^{-1}}$ and $3.125 \times 10^{-3}\ \mathrm{mol\ L^{-1}s^{-1}}$

Q79. Single correct

Which one of the following statements for the order of a reaction is incorrect?

(A) Order can be determined only experimentally

(B) Order is not influenced by stoichiometric coefficient of the reactants

(C) Order of reaction is sum of power to the concentration terms of reactants to express the rate of reaction

(D) Order of reaction is always whole number ✓

Q80. Single correct

The half-life of a substance in a certain enzyme-catalysed first order reaction is 138 s. The time required for the concentration of the substance to fall from 1.28 mg L^{-1} to 0.04 mg L^{-1}:

(A) 276 s

(B) 414 s

(C) 552 s

(D) 690 s ✓

Q81. Single correct

In a zero-order reaction for every 10° rise of temperature, the rate is doubled. If the temperature is increased from 10°C to 100°C, the rate of the reaction will become:

(A) 64 times

(B) 128 times

(C) 256 times

(D) 512 times ✓

Q82. Single correct

A reaction having equal energies of activation for forward and reverse reactions has:

(A) $\Delta H = \Delta G = \Delta S = 0$

(B) $\Delta S = 0$

(C) $\Delta G = 0$

(D) $\Delta H = 0$ ✓

Q83. Single correct

The activation energy of a reaction can be determined from the slope of which of the following graphs?

(A) $\frac{\ln k}{T}$ v/s. T

(B) $\ln k$ v/s. $\frac{1}{T}$ ✓

(C) $\frac{T}{\ln k}$ v/s. $\frac{1}{T}$

(D) $\ln k$ v/s. T

Q84. Single correct

The rate constant of the reaction A → B is 0.6×10^{-3} $\mathrm{M\ sec^{-1}}$. If the concentration of A is 5 M, then concentration of B after 20 min is:

(A) 0.36 M

(B) 0.72 M ✓

(C) 1.08 M

(D) 3.60 M

Q85. Single correct

The decomposition of phosphine $(\mathrm{PH} * 3)$ on tungsten at low pressure is a first-order reaction. It is because the

(A) rate is independent of the surface coverage

(B) rate of decomposition is very slow

(C) rate is proportional to the surface coverage ✓

(D) rate is inversely proportional to the surface coverage

Q86. Single correct

A first order reaction has a specific reaction rate of $10^{-2}\ \mathrm{s}^{-1}$. How much time will it take for 20 g of the reactant to reduce to 5 g?

(A) 138.6 s ✓

(B) 346.5 s

(C) 693.0 s

(D) 238.6 s

Q87. Single correct

When initial concentration of the reactant is doubled, the half-life period of a zero-order reaction

(A) is halved

(B) is doubled ✓

(C) is tripled

(D) remains unchanged

Q88. Single correct

If the rate constant for a first order reaction is k, the time (t) required for the completion of 99% of the reaction is given by:

(A) $t = 0.693/k$

(B) $t = 6.909/k$

(C) $t = 4.606/k$ ✓

(D) $t = 2.303/k$

Q89. Single correct

A first order reaction has a rate constant of $2.303 \times 10^{-3}\ \mathrm{s}^{-1}$. The time required for 40g of this reactant to reduce to 10 g will be [Given that $\log *102 = 0.3010$]

(A) 230.3 s

(B) 301 s

(C) 2000 s

(D) 602 s ✓

Q90. Single correct

For a reaction, activation energy $E_a = 0$ and the rate constant at 200K is $1.6 \times 10^6 \mathrm{s}^{-1}$. The rate constant at 400K will be [Given: gas constant $R = 8.314\ \mathrm{J\ K}^{-1}\mathrm{mol}^{-1}$]

(A) $3.2 \times 10^4\ \mathrm{s}^{-1}$

(B) $1.6 \times 10^6\ \mathrm{s}^{-1}$ ✓

(C) $1.6 \times 10^3\ \mathrm{s}^{-1}$

(D) $3.2 \times 10^6\ \mathrm{s}^{-1}$

Q91. Single correct

Which of the following statement is correct.

(A) The direct elongation of the radicle leads to formation of adventitious root

(B) Stem of maize produces roots from the lower nodes called as climbing root.

(C) A bud is present in the axile of petiole in both simple and compound leaves, but not in the axile of leaflets of the compound leaf. ✓

(D) In pinnately compound leaf, incisions of leaf are directed from leaf margins to apex of the petiole and all leaflets are attached at a common point.

Q92. Single correct

Match the following :-

Column-A		Column-B	
(A)	Whorled phyllotaxy	(i)	Pea
(B)	Perigynous	(ii)	Rose, Plum and Peach
(C)	Phyllode	(iii)	*Alstonia*
(D)	Papilionaceous petals	(iv)	*Australian Acacia*

(A) A-iii, B-ii, C-iv, D-i ✓

(B) A-i, B-iii, C-ii, D-iv

(C) A-iv, B-ii, C-i, D-ii

(D) A-iii, B-ii, C-i, D-iv

Q93. Single correct

Which of the following is incorrect :-

(A) In Rose, Peach and Plum, gynoecium is situated in the centre and other parts of the flower are located on the rim of the thalamus almost at the same level

(B) A lateral branch with short internodes and each node bearing a rosette of leaves and a tuft of roots is found in aquatic plants like eichhornia and pistia

(C) All the leaflets are found attached at a common point of rachis in pinnately compound leaves ✓

(D) In Cassia and Gulmohur, Sepals or petals overlap one another but not in any particular direction

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Q94. Single correct

Dicot plants differ from monocots in many characters like :-

(A) Two cotyledons in seeds while in monocots, they are either 3 or multiple of three.

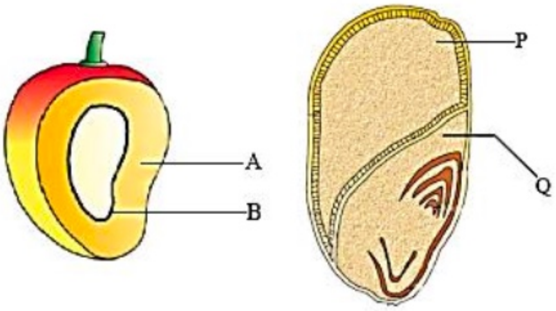
(B) More number of pollengrains in stamen as compare to monocot.

(C) Parallel venation in dicot while reticulate in monocot.

(D) Flowers are tetra or pentamerous while trimerous in monocots. ✓

Q95. Single correct

Identify the parts labelled A,B and P,Q in the figure given below and select the right option :-



	A	B	P	Q
(1)	Epicarp	Mesocarp	Scutellum	Endosperm
(2)	Mesocarp	Endocarp	Endosperm	Scutellum
(3)	Aleurone layer	Endocarp	Mesocarp	Radicle
(4)	Mesocarp	Seed	Endosperm	Plumule

(A) 1

(B) 2 ✓

(C) 3

(D) 4

Q96. Single correct

Which of the following statement is correct :-

(A) In tomato fruit is a capsule

(B) Seeds of orchids have oily endosperm

(C) Placentation in Dianthus is free central ✓

(D) Tetradynamous condition is found in pea

Q97. Single correct

Which of the following is not true for a dicotyledonae plant ?

(A) Bear seeds with two cotyledon

(B) Possess fibrous root system ✓

(C) Possess reticulate venation in leaves

(D) May have adventitious roots in some members

Q98. Single correct

In chinarose :-
(A) Stamens are monoadelphous and aestivation is twisted
(B) Placentation is axile and anthers are monothecous and bisporangiate
(C) Staminal tube is found attached with petals
(D) Flowers are pentamerous actinomorphic with superior ovary
(E) Pentacarpellary gynoecium exhibits five free stigma
(F) Sepals are gamosepalous whereas petals are polypetalous

(A) A,B and F

(B) B and D

(C) C,A and E

(D) All are correct ✓

Q99. Single correct

A diagnostic trait for identification of fabaceae flower is :-

(A) Axile placentation

(B) Vexillary aestivation ✓

(C) Cruciform corolla

(D) Epipetalous

Q100. Single correct

Which of the following is not related to solanaceae.

(A) Axile placentation

(B) Dicot family

(C) Racemose inflorescence ✓

(D) Solitary inflorescence

Q101. Single correct

Axile Placentation is found in :-

(A) Compositae

(B) Fabaceae

(C) Cruciferae

(D) Liliaceae ✓

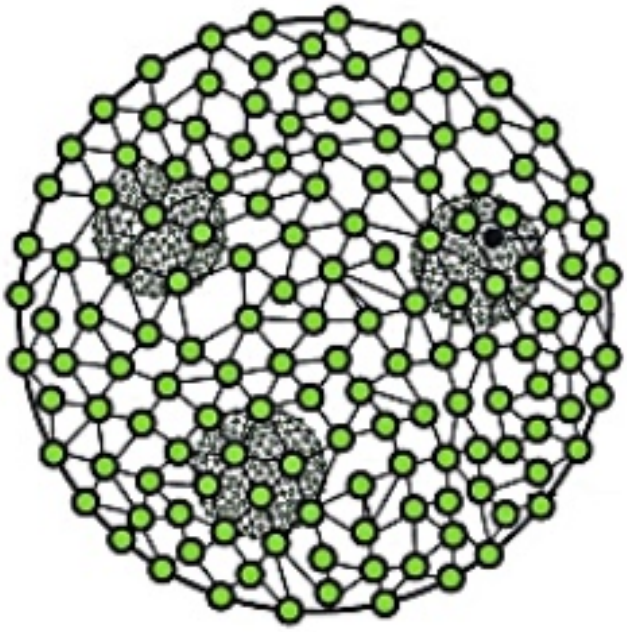
Q102. Single correct

Observe the given algae and select the correct option in which all answers are correct of given questions :-

(a) Name of algae ?

(b) Type of sexual reproduction ?

(c) Name of class ?



(A) Laminaria : Isogamous : Chlorophyceae

(B) Volvox : Oogamous : Chlorophyceae ✓

(C) Fucus : Isogamous : Rhodophyceae

(D) Chara : Oogamous : Rhodophyceae

Q103. Single correct

Select false statement ?

(A) The members of phaeophyceae are found primarily in marine habitats.

(B) The members of chlorophyceae are usually grass green due to the dominance of pigments chlorophyll a and chlorophyll b.

(C) Majority of red algae are marine with greater concentration found in warmer areas. ✓

(D) Red algae reproduce by zoospores formation.

Q104. Single correct

Stored food of red algae is similar to :-

(A) Chitin

(B) Mannitol

(C) Amylopectin and glycogen ✓

(D) Glycogen and chitin

Q105. Single correct

Pyrenoids are made up of –

(A) Core of starch surrounded by sheath of protein

(B) Core of protein surrounded by fatty sheath

(C) Proteinaceous centre and starchy sheath ✓

(D) Core of nucleic acid surrounded by protein

Q106. Single correct

In brown algae asexual spores are :-

(A) Pear-shaped and have two unequal flagella ✓

(B) Pear-shaped and have two unequal cilia

(C) Oval-shaped and have two unequal flagella

(D) Comma-shaped and biflagellate

Q107. Single correct

In the member of phaeophyceae which part of body functions as photosynthetic part :-

(A) Vellum

(B) Stipe

(C) Hold fast

(D) Frond ✓

Q108. Single correct

Choose the correct match out of the following-

(A) Phaeophyceae - Fucoxanthin, Starch

(B) Rhodophyceae - Phycocyanin, Laminaria

(C) Chlorophyceae - Chlorophyll, Starch ✓

(D) Cyanophyceae - Phycoerythrin, Floridean Starch

Q109. Single correct

The members of rhodophyceae are commonly called red algae because of the :-

(A) they are found in greater concentrations of warmer area of Ocean.

(B) they have floridean starch as a stored food.

(C) prodominance of r-phycoerythrin in their body. ✓

(D) they show complex post fertilization developments.

Q110. Single correct

Which option is not related with phaeophyceae?

(A) Gelatinous coating of algin.

(B) Gametes are pear shaped and bear two laterally attached flagella.

(C) The food is stored as floridean starch. ✓

(D) They possess chlorophyll a,c, carotenoids and xanthophylls.

Q111. Single correct

The members of rhodophyceae commonly called red algae because of the

(A) They are found in greater concentrations of warmer area of ocean

(B) They have floridean starch as a stored food

(C) Prodominance of r-phycoerythrin in their body ✓

(D) They show complex post fertilization developments

Q112. Single correct

Red algae differ from the green algae and brown algae in having :-

(A) Chlorophyll 'a'

(B) Haemoglobin with in their cells

(C) Aquatic nature

(D) No motile stage present in their life cycle ✓

Q113. Single correct

Kelps belongs to :-

(A) Chlorophyceae

(B) Pheophyceae ✓

(C) Rhodophyceae

(D) Cyanophyceae

Q114. Single correct

Sex organs of algae are usualy

(A) Multicellular

(B) Unicellular ✓

(C) Unicellular and Jacketed

(D) Multicellular and Jacketed

Q115. Single correct

Study the following table carefully and give the correct answer from the options :-

	Column-I	Column-II	Column-III
(i)	*Chlorella*	Protein source	Cyanobacteria
(ii)	*Spirullina*	Chlorophyceae	Protein source
(iii)	Halophiles	Grow in salty areas	Archaebacteria
(iv)	Saprophytic protists	Slime moulds	Forms plasmodium

(A) i & ii - incorrect, iii & iv correct ✓

(B) i & iii - incorrect, ii & iv correct

(C) ii & and iv - incorrect, iii & i - correct

(D) i, ii, iii - correct, only iv is incorrect

Q116. Single correct

Find the correct statement :-

(A) Volvox and Fucus show isogamy whereas Ulothrix and Spirogyra show oogamy

(B) Red algae have a dominant pigment r-phycocyanin

(C) Brown algae produce gametes with two equal lateral flagella

(D) Agar is obtained from Gelidium and Gracilaria ✓

Q117. Single correct

Which of the following structure is formed in the life cycle of Funaria between spore and main plant body ?

(A) Prothallus

(B) Embryo

(C) Protonema

(D) Spore mother cell ✓

Q118. Single correct

Which of the following statements are true about Marchantia :-

(A) Thallus is liver shaped

(B) Dorsal surface of thallus bears gemma cups and ventral surface of thallus bears rhizoids & scales

(C) Sporophyte is made up of foot, seta & capsule

(D) All the above ✓

Q119. Single correct

Which one of the following is not a ecological importance of moss plants :-

(A) Some mosses provide food for herbaceous mammals birds and other animals

(B) Very high water holding capacity of mosses is useful for trans-shipment of living materials ✓

(C) Mosses along with lichens are the pioneering organism to colonise rocks

(D) Mosses form dense mats on the soil and reduce the impact of falling rain

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Q120. Single correct

Which is the correct statement for bryophytes?

(A) Their sporophytic generation is smaller than gametophytic and is generally parasite ✓

(B) Meiosis takes place in reproductive organs and results in the formation of gametes

(C) Their roots function, for both conduction and support.

(D) Their gametophytic generation is the dominant phase, which produces spores and each spore gives rise to a new gametophytic plant.

Q121. Single correct

With respect to fungal sexual cycle, choose the correct sequence of events :-

(A) Karyogamy, Plasmogamy & Meiosis

(B) Meiosis, Plasmogamy & Karyogamy

(C) Plasmogamy, Karyogamy & Meiosis ✓

(D) Meiosis, Karyogamy & Plasmogamy

Q122. Single correct

Gemmae are :-

(A) Asexual reproductive structure of Moss

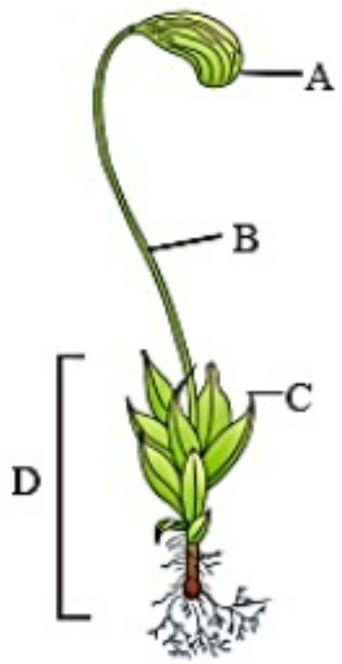
(B) Sexual reproductive structure of Liverwort.

(C) Sexual reproductive structure of Moss

(D) Asexual reproductive structure. ✓

Q123. Single correct

Given below is the diagram of organism identify the parts labelled A, B, C and D, and select the right option about them :-



	Part (A)	Part (B)	Part (C)	Part (D)
1	Sporophyte	Stalk	Leaves	Foot
2	Capsule	Foot	Leaves	Gametophyte
3	Sporophyte	Stalk	Foot	Gametophyte
4	Capsule	Seta	Leaves	Gametophyte

(A) 1

(B) 2

(C) 3

(D) 4 ✓

Q124. Single correct

Arrange the following events of life cycle of bryophytes in correct sequence :-

I. Germination of spore
II. Gametes formation
III. Formation of Gametophyte
IV. Fertilization
V. Embryo formation

(A) I,III,IV,V,II

(B) II,IV,I,V,III

(C) III,II,IV,V,I ✓

(D) V,I,II,III,IV

Q125. Single correct

Which of the following is called as peat moss:-

(A) Funaria

(B) Sphagnum ✓

(C) Polytrichum

(D) Riccia

Q126. Single correct

Select the incorrect statement from the followings about bryophytes.

(A) The plant Body of liverworts is not thalloid and made of rhizoids, stem and leaves. ✓

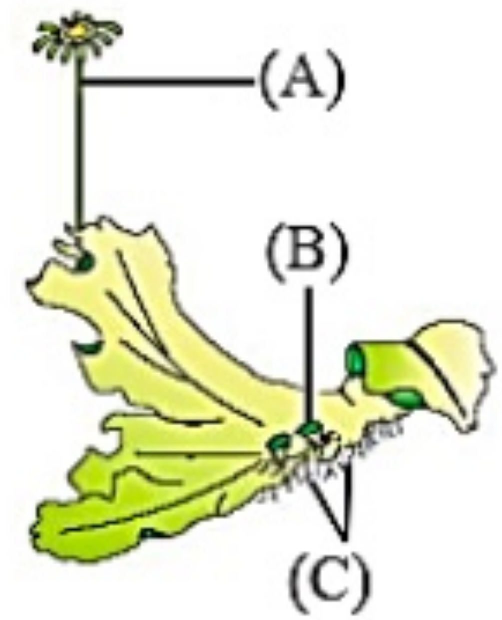
(B) Mosses have upright, slender axis bearing spirally arranged leaves.

(C) Spores germinate to form gametophyte

(D) The zygote produces a sporophyte

Q127. Single correct

In the labelled figure of Marchantia, what are A, B and C.



(A) Archegoniophore, Gemma cup, Rhizoids ✓

(B) Rhizoids, Archegoniophore, Gemma cup

(C) Gemma cup, Archegoniophore, Rhizoids

(D) Rhizoids, Antheridiophore, Gemma cup

Q128. Single correct

The life cycle in Bryophyta is haplo - diplontic type, it means that -

(A) Haploid generation is multicellular and diploid generation is single celled.

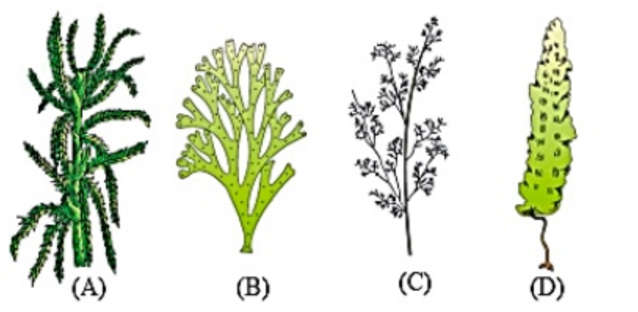
(B) Haploid generation is dominant and multicellular, diploid generation is also multicellular but comparatively less developed. ✓

(C) Haploid and diploid generations are equally developed.

(D) Haploid generation is highly reduced.

Q129. Single correct

Identify the given diagrams and select the correct options :-



	A	B	C	D
(1)	*Sphagnum*	*Porphyra*	*Dictyota*	*Lamineria*
(2)	*Sphagnum*	*Dictyota*	*Polysiphonia*	*Lamineria*
(3)	*Sphagnum*	*Fucus*	*Lamineria*	*Chara*
(4)	*Sphagnum*	*Chara*	*Polysiphonia*	*Lamineria*

(A) 1

(B) 2 ✓

(C) 3

(D) 4

Q130. Single correct

Where does reduction division take place in the life cycle of a fern

(A) Capsule

(B) Ovule

(C) Pollen sac

(D) Sporangium ✓

Q131. Single correct

In pteridophytes, spores germinate to give rise to inconspicuous, small but multicellular, free living, mostly photosynthetic thalloid gametophytes, called :-

(A) Protonema

(B) Prothallus ✓

(C) Endosperm

(D) Embryosac

Q132. Single correct

Which of the following pair is an example of heterosporous pteridophyte ?

(A) Psilotum and Adiantum

(B) Salvinia and Selaginella ✓

(C) Lycopodium and Dryopteris

(D) Nephrolepis and Adiantum

Q133. Single correct

Arrange the following events in a correct sequence with respect to life cycle of fern :-
(I) Fertilization
(II) Liberation of spores
(III) Prothallus
(IV) Embryo

(A) II, III, I, IV ✓

(B) IV, III, II, I

(C) I, II, III, IV

(D) I, IV, III, II

Q134. Single correct

Which one of the following is not related with heterosporous pteridophytes?

(A) Show precursor to the seed habit

(B) Produce monoecious gametophyte ✓

(C) Gametophyte retained on the parent sporophyte for variable periods.

(D) Selaginella and Salvinia

Q135. Single correct

The gametophyte of pteridophytes requires to grow-

(A) Warm, damp and shady place

(B) Cool, damp and shady place ✓

(C) Warm, dry and shady place

(D) Cool, dry and place of well sunshine

Q136. Single correct

A student makes the following statements about the structure of proteins.
(A) The primary structure of a protein is determined by the sequence of amino acids, which in turn is dictated by the sequence of nucleotides in the mRNA.
(B) A right-handed helical structure (α-helix) is an example of the tertiary structure, stabilized by hydrogen bonds between the C=O of one amino acid and the N-H of the next fourth amino acid.
(C) Adult human haemoglobin exhibits quaternary structure because it is an assembly of four polypeptide subunits.
(D) Denaturation of a protein by heat or pH changes disrupts the primary structure by breaking peptide bonds.
How many of the above statements are correct?

(A) One

(B) Two ✓

(C) Three

(D) Four

Q137. Single correct

Consider the kinetics of an enzymatic reaction.
Which of the following statements are incorrect?
(A) Vmax is the maximum velocity of a reaction, which is reached when the enzyme molecules are fully saturated with substrate.
(B) Km is a constant that indicates the substrate concentration at which the reaction proceeds at full velocity (Vmax).
(C) A competitive inhibitor structurally resembles the substrate and binds irreversibly to the active site.
(D) The presence of a non-competitive inhibitor decreases the apparent Vmax but does not change the Km of the enzyme.
Select the option with all incorrect statements.

(A) B and C only ✓

(B) A and D only

(C) B, C, and D only

(D) A, B, and C only

Q138. Single correct

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): In a DNA molecule, the two polynucleotide chains are antiparallel.

Reason (R): The phosphodiester bonds in one strand go from 3' to 5', while in the other strand, they go from 5' to 3'.

In the light of the above statements, choose the correct answer from the options given below:

(A) A is true but R is false.

(B) A is false but R is true.

(C) Both A and R are true and R is the correct explanation of A. ✓

(D) Both A and R are true but R is NOT the correct explanation of A.

Q139. Single correct

An unknown substance is added to a solution containing the enzyme succinate dehydrogenase and its substrate, succinate. The reaction rate decreases. Upon adding more succinate, the reaction rate returns to normal. The unknown substance is likely:

(A) a non-competitive inhibitor.

(B) a denaturing agent like high temperature.

(C) malonate, a competitive inhibitor. ✓

(D) an allosteric activator.

Q140. Single correct

Which of the following R groups, when attached to the α-carbon of an amino acid, would confer basic properties to the molecule?

(A) -H

(B) $-CH_2-COOH$

(C) $-(CH_2)_4-NH_2$ ✓

(D) $-CH_2-OH$

Q141. Single correct

A patient is diagnosed with a disease where their cells cannot effectively form phosphodiester bonds. This would most directly inhibit the synthesis of:

(A) Polypeptides and Triglycerides

(B) Starch and Glycogen

(C) DNA and RNA ✓

(D) Phospholipids and Cholesterol

Q142. Single correct

During the analysis of the chemical composition of living tissue, what is the fundamental difference between the “ash” and the acid-insoluble fraction?

(A) Ash contains inorganic elements, while the acid-insoluble fraction contains organic polymers. ✓

(B) Ash represents all carbon compounds, while the acid-insoluble fraction is only proteins.

(C) Ash contains micromolecules, while the acid-insoluble fraction contains macromolecules.

(D) Ash is the fraction burnt away, while the acid-insoluble fraction remains as a pellet.

Q143. Single correct

Match the List-I with List-II.

Choose the correct answer from the options given below:

	List-I		List-II
(A)	Oxidoreductase	(I)	Linking together of 2 compounds
(B)	Lyase	(II)	Inter-conversion of isomers
(C)	Ligase	(III)	Removal of groups from substrates by mechanisms other than hydrolysis leaving double bonds
(D)	Isomerase	(IV)	Catalyze oxido-reduction between two substrates

(A) A-II, B-III, C-IV, D-I

(B) A-IV, B-III, C-I, D-II ✓

(C) A-III, B-I, C-II, D-IV

(D) A-III, B-II, C-IV, D-I

Q144. Single correct

A researcher is studying different polysaccharides. Which of the following statements would be correct?
(A) Inulin is a polymer of fructose and is used to test kidney function.
(B) Chitin is a homopolymer of N-acetylglucosamine, found in the exoskeleton of arthropods.
(C) Glycogen is a branched polymer of glucose stored in the liver and muscles of animals.
(D) β-glucose linked by β 1-4 glycosidic bonds.

(A) A and C only

(B) B, C, and D only

(C) A, B, and C only

(D) A, B, C, and D ✓

Q145. Single correct

Evaluate the following statements regarding lipids and select the correct set.
(A) Gingelly oil has a lower melting point compared to butter, hence it remains as an oil in winters.
(B) Lecithin is an example of a phospholipid found in the cell membrane and is a diglyceride.
(C) Cholesterol, a sterol lipid, is the precursor for steroid hormones like testosterone and also contributes to membrane fluidity.
(D) Palmitic acid has 16 carbon atoms including the carboxyl carbon, while arachidonic acid has 20 carbon atoms excluding the carboxyl carbon.
Choose the correct option:

(A) A and C only

(B) A, B, and C only ✓

(C) B and D only

(D) A, B, C, and D

Q146. Single correct

Which of the following would be completely removed as a gas during the process of burning the tissue?

(A) Calcium (Ca)

(B) Sulphate (SO_4^{2-})

(C) Sodium (Na^+)

(D) Carbon (from organic compounds) ✓

Q147. Single correct

Read the statements related to protein structure and function.
(A) The sequence of amino acids in a protein's primary structure is responsible for its unique folding and ultimate function.
(B) An α-helix is a coiled secondary structure where every peptide bond in the backbone participates in hydrogen bonding.
(C) The tertiary structure represents the overall 3D shape of a polypeptide, involving interactions between the R-groups of amino acids that may be far apart in the primary sequence.
(D) The function of GLUT-4 is to act as an intercellular ground substance, enabling cell-to-cell communication.
Select the option containing all correct statements.

(A) A and C only

(B) A, B, and C only ✓

(C) B and D only

(D) A, B, C, and D

Q148. Single correct

Evaluate the following comparisons between different biomolecules.
(A) Chitin and Cellulose are both structural homopolymers, but chitin contains nitrogen while cellulose does not.
(B) Adenylic acid is a nucleotide, whereas adenosine is a nucleoside; the difference is a single phosphate group.
(C) Both haemoglobin and RuBisCO have quaternary structures, but haemoglobin is the most abundant protein in the animal world, while RuBisCO is the most abundant in the biosphere.
(D) Saturated fatty acids have higher melting points than unsaturated fatty acids of the same carbon chain length because the absence of double bonds allows for more compact packing.
How many of the above comparisons are correct?

(A) One

(B) Two

(C) Three ✓

(D) Four

Q149. Single correct

A student prepares a list of statements about lipids. Identify the correct ones.
(A) Lipids are polymers of fatty acids and glycerol.
(B) Because lipids have a molecular weight of less than 1000 Da, they are technically not biomacromolecules.
(C) Phospholipids are amphipathic molecules, with a hydrophilic head and two hydrophobic tails, which is essential for forming the lipid bilayer of membranes.
(D) Cholesterol is a derived lipid that is hydrophilic and easily soluble in blood plasma.
Choose the correct option.

(A) B and C only ✓

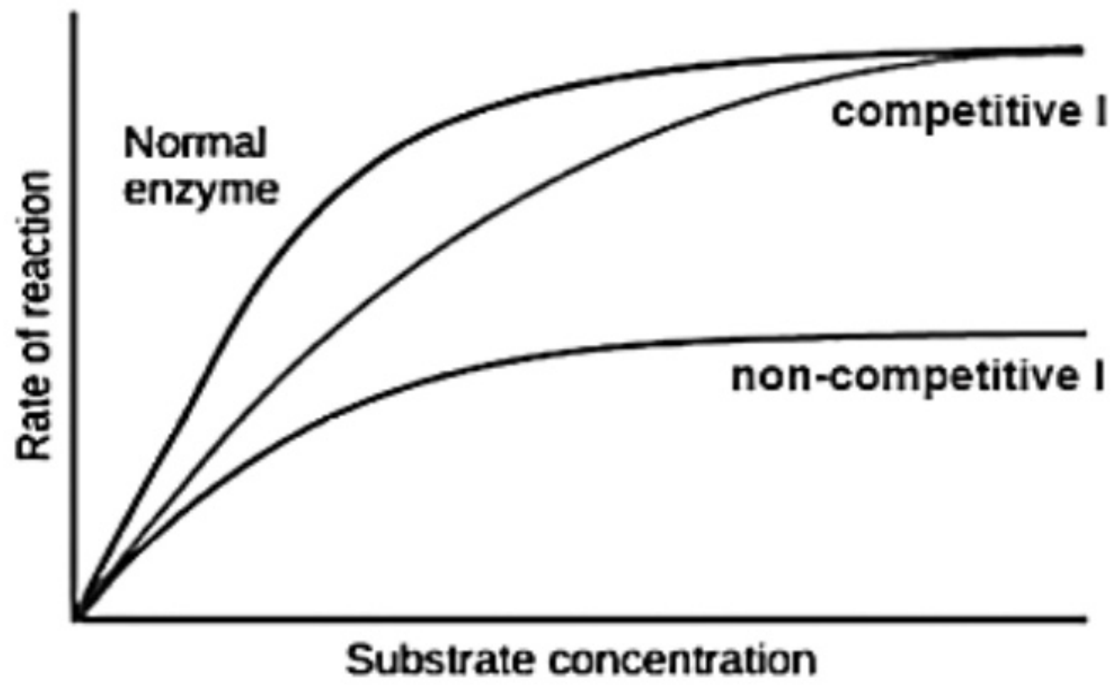
(B) A and D only

(C) A, B, and C only

(D) B, C, and D only

Q150. Single correct

The graph below shows the effect of an inhibitor on an enzyme-catalyzed reaction. Based on the graph, what can you conclude about the inhibitor and its effects?



(A) It is a competitive inhibitor because it increases the Km but does not change the Vmax.

(B) It is a non-competitive inhibitor because it decreases the Vmax but does not change the Km. ✓

(C) It is a competitive inhibitor because it decreases the Vmax.

(D) It is a non-competitive inhibitor because it increases the affinity of the enzyme for the substrate.

Q151. Single correct

When a living tissue is ground with trichloroacetic acid, two fractions are obtained: acid-soluble and acid-insoluble. Which of the following is an incorrect statement regarding this process?

(A) The acid-soluble pool contains micromolecules like amino acids and simple sugars.

(B) The acid-insoluble fraction contains macromolecules like proteins and nucleic acids.

(C) Lipids, despite being micromolecules, are found in the acid-insoluble fraction.

(D) The acid-soluble fraction represents the exact chemical composition of the cytoplasm, while the insoluble fraction represents the organelles. ✓

Q152. Single correct

Consider the classification of enzymes. Which of the following is correctly matched?

(A) Enzyme catalyzing A-B + H_2O → A-OH + B-H --- Hydrolase

(B) Enzyme catalyzing A + B → A-B --- Ligase

(C) Enzyme catalyzing conversion of Glucose to Fructose --- Isomerase

(D) Enzyme catalyzing S + S' → S-red + S'-ox --- Oxidoreductase

(A) A, C, and D only

(B) A and B only

(C) A, B, C, and D ✓

(D) C and D only

Q153. Single correct

Match List-I with List-II:

Choose the correct answer from the options given below:

	List-I (Metabolite)		List-II (Class)
A.	Vinblastin	I.	Alkaloid
B.	Ricin	II.	Terpenoid
C.	Monoterpenes	III.	Toxin
D.	Morphine	IV.	Drug

(A) A-IV, B-III, C-II, D-I ✓

(B) A-III, B-IV, C-I, D-II

(C) A-IV, B-II, C-III, D-I

(D) A-I, B-III, C-IV, D-III

Q154. Single correct

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): The tertiary structure of a protein is absolutely necessary for its biological activity.

Reason (R): The active site of an enzyme is formed by specific amino acid residues that are brought together by the folding of the polypeptide chain into its tertiary structure.

In the light of the above statements, choose the correct answer from the options given below:

(A) A is true but R is false.

(B) A is false but R is true.

(C) Both A and R are true and R is the correct explanation of A. ✓

(D) Both A and R are true but R is NOT the correct explanation of A.

Q155. Single correct

How many of the following biomolecules are considered secondary metabolites?

Insulin, Ricin, Lecithin, Codeine, Curcumin, Valine, Vinblastin

(A) Three

(B) Four ✓

(C) Five

(D) Six

Q156. Single correct

Name the four elements called ‘Big four’, which make up 95% of all elements found in a living system.

(A) C, H, O, P

(B) C, H, O, N ✓

(C) C, N, O, K

(D) C, H, O, S

Q157. Single correct

Select the secondary metabolites from the list given below.

I. Alkaloids
II. Flavonoids
III. Rubber
IV. Essential oils
V. Antibiotics
VI. Coloured pigments
VII. Scents
VIII. Gums
IX. Spices

Choose the correct option.

(A) All except I and VII

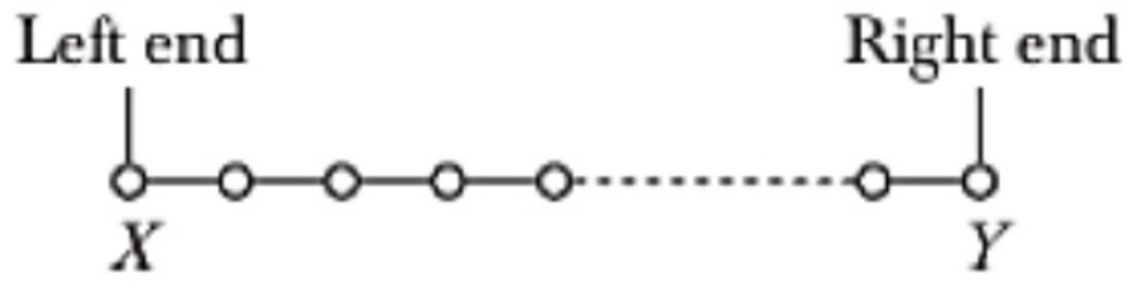
(B) All except II and IX

(C) I, III, IV and VI

(D) I to IX ✓

Q158. Single correct

Name the term given to the left and right ends of a polysaccharide.



(A) Left end–N-terminal, Right end–C-terminal end

(B) Left end–Reducing, Right end–non-reducing end

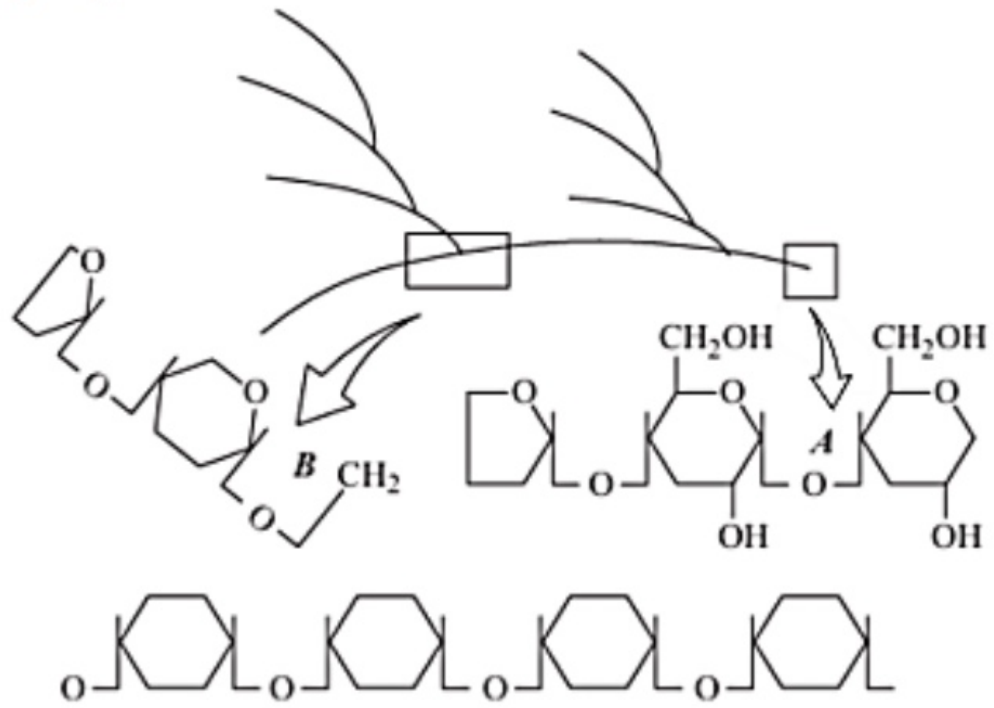
(C) Left end–non-reducing, Right end–Reducing end ✓

(D) Left end–C-terminal, Right end–N-terminal end

Q159.

Single correct

Identify A and B bonds in the following diagrammatic representation of a portion of glycogen. Choose the correct option.



(A) A = 1,6 α-glycosidic bonds, B=1,4 α-glycosidic bonds

(B) A = 1,1 α-glycosidic bonds, B=1,1 α-glycosidic bonds

(C) A = 1,4 α-glycosidic bonds, B=1,4 α-glycosidic bonds

(D) A = 1,4 α-glycosidic bonds, B=1,6 α-glycosidic bonds ✓

Q160. Single correct

The diagram shows a metabolic pathway:
What would happen to the rate of production of ‘D’, if enzyme 1 was not present?

$$A \xrightarrow{\text{Enzyme–1}} B \xrightarrow{\text{Enzyme–2}} C \xrightarrow{\text{Enzyme–3}} D$$

(A) It would stop ✓

(B) It would be increased

(C) It would be decreased

(D) No effect

Q161. Single correct

A protein having both structural and enzymatic properties is:

(A) collagen

(B) trypsin

(C) myosin ✓

(D) actin

Q162. Single correct

Which enzyme is used by the biscuit manufactures to lower the protein level of flour?

(A) Amylases

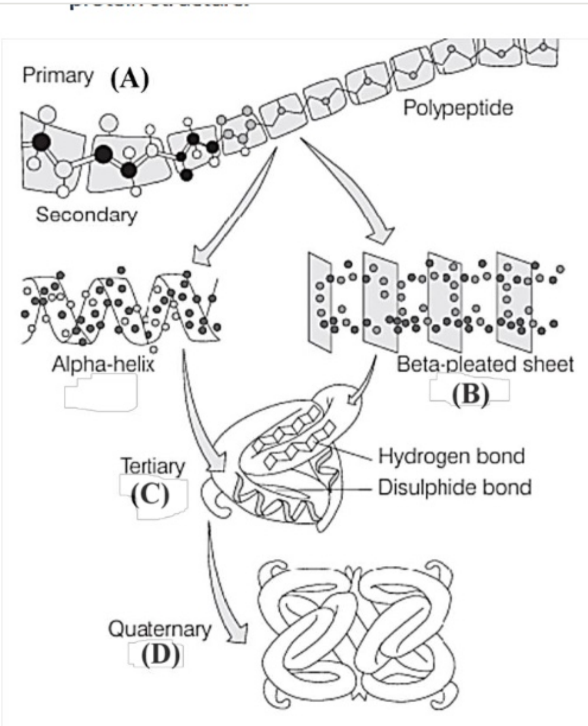
(B) Cellulases

(C) Proteases ✓

(D) Xylases

Q163. Single correct

Refer to the given figure of various levels of protein structure.



(A) Structure A-H-bonding in single amino acid chain only, Structure B-H-bonding in between two or more polypeptide chains, Structure C-Further coiling in α-helix only, Structure D-Both α-helix and β-sheet joined and coiled together

(B) Structure A-Peptide bond in between two or more polypeptide chains, Structure B-H-bonding in single amino acid chain only, Structure C-Further coiling in α-helix only, Structure D- Both α-helix and β-sheet joined and coiled together

(C) Structure A-Peptide bond in between two or more polypeptide chains, Structure B-H-bonding in single amino acid chain only, Structure C-Both α-helix and β-sheet joined and coiled together, Structure D- Further coiling in α-helix only

(D) Structure A-Peptide bond in single amino acid chain only, Structure B-H-bonding in between stretches of a polypeptide chains, Structure C-Further folding of two or more secondary structures, Structure D- Further folding of a number of tertiary structures ✓

Q164. Single correct

Match List-I With List-II.
Choose the correct answer from the options given below:

List-I (Enzymes)		List-II (Characteristics)	
(A)	Dehydrogenases	(I)	Interconversion of optical, geometrical, positional isomers
(B)	Ligases	(II)	Linking together of two bonds
(C)	Isomerases	(III)	Oxidoreduction between two substrates
(D)	Hydrolases	(IV)	Hydrolysis of bonds

(A) A-I, B-II, C-III, D-IV

(B) A-III, B-II, C-I, D-IV ✓

(C) A-IV, B-III, C-II, D-I

(D) A-II, B-I, C-IV, D-III

Q165. Single correct

Which of the following processes occurs without the involvement of an enzyme in a biological system?

(A) CO_2 dissolving in water ✓

(B) Separation of DNA strands

(C) Breakdown of sucrose

(D) Peptide bond synthesis

Q166. Single correct

Consider the following statements.

I. Glycogen is a homopolymer whereas proteins are heteropolymers.

II. Nucleotide is an acid insoluble compound formed by phosphorylation of nucleoside.

III. Water is the most abundant chemical in living organism.

IV. A complete catalytic active enzyme with its bound prosthetic group is called apoenzyme

V. Haemoglobin molecule is composed of four different polypeptide chains – two α-chains and two β-chains.

Which of the above statement is/are incorrect?

(A) I, II and V only

(B) II, III and IV only

(C) II and IV only

(D) II, IV and V only ✓

Q167. Single correct

How many of the following statements are correct?

I. About 98% of mass of every living organism is composed of just six elements including C, H, O, N and phosphorus and sulphur.

II. Chitin is a homopolymer of N-acetyl glucosamine.

III. Glycine is a sulphur containing amino acid.

IV. Arachidonic acid has 20 carbon atoms including the carboxyl carbon.

V. Oils have a lower melting point.

VI. The neural tissues have lipids with more complex structures.

(A) Three

(B) Four

(C) Five ✓

(D) Six

Q168. Single correct

Match List-I with List-II.

Choose the correct answer from the options given below:

List-I		List-II	
(A)	Non reducing sugar	(I)	Used in Renal tests
(B)	Inulin	(II)	Tertiary structure
(C)	Fibroin	(III)	Sucrose
(D)	Deoxyribose	(IV)	Present in DNA

(A) A-III, B-I, C-II, D-IV ✓

(B) A-II, B-I, C-III, D-IV

(C) A-III, B-I, C-IV, D-II

(D) A-IV, B-II, C-I, D-III

Q169. Single correct

In competitive inhibition, which of the following is true?

(A) $E + I \rightleftharpoons EI$ ✓

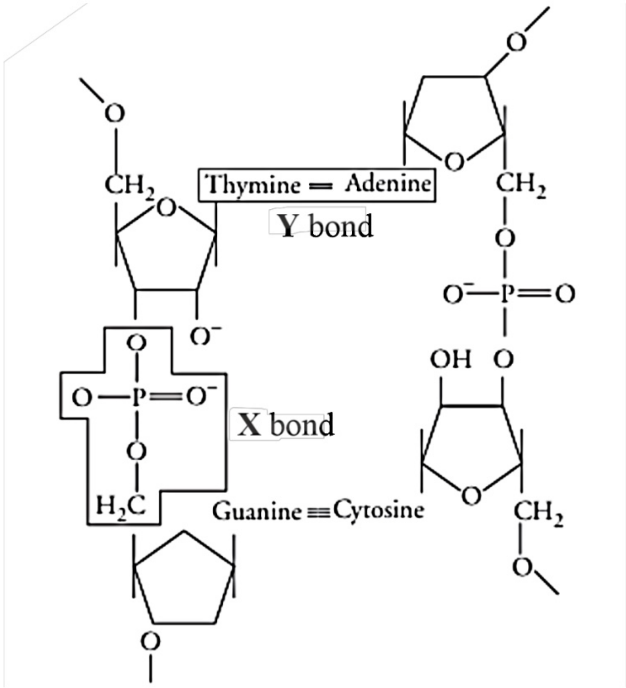
(B) $E + I \rightleftharpoons EI + S \rightleftharpoons EIS$

(C) $S + I \rightleftharpoons SI$

(D) $ES + I \rightleftharpoons ESI$

Q170. Single correct

Which bonds are indicated by X and Y in the given diagram?



(A) X–Glycosidic bond, Y–Hydrogen bond

(B) X–Phosphodiester bond, Y–Hydrogen bond ✓

(C) X–Glycosidic bond, Y–Phosphodiester bond

(D) X–Phosphodiester bond, Y–Glycosidic bond

Q171. Single correct

In the enzyme which catalyses the breakdown of:

$H_2O_2 \rightarrow H_2O + O_2$

The prosthetic group is:

(A) Nicotinamide adenine dinucleotide

(B) Haem ✓

(C) Zinc

(D) Niacin

Q172. Single correct

Read the following statements.
I. Nucleic acids carry hereditary information and are passed on from parental generation to progeny.
II. Adhesives such as fevicol are also a type of natural gum, similar in terms of source and composition.
III. Qualitative tests are used for detecting proteins, fats/oils and amino acids in biological samples.
IV. Apart from a lot of biological functions proteins like collagen and elastin are also used in cosmetic industry.
V. Glycolic acid is one of the essential amino acids.
Which of the above statement is/are incorrect?

(A) I, II and V only

(B) II, III and IV only

(C) II and IV only

(D) II and V only ✓

Q173. Single correct

Choose the correct statement.

(A) Enzymes start a chemical reaction but do not change its equilibrium.

(B) Digestive enzymes perform intracellular catalysis.

(C) Enzymes are mainly functional inside the living cells but can be made to catalyse reactions outside living cells. ✓

(D) All enzymes are proteinaceous in nature with no exceptions.

Q174. Single correct

Choose the correct statement.

(A) Active site is responsible for attracting and holding substrate. ✓

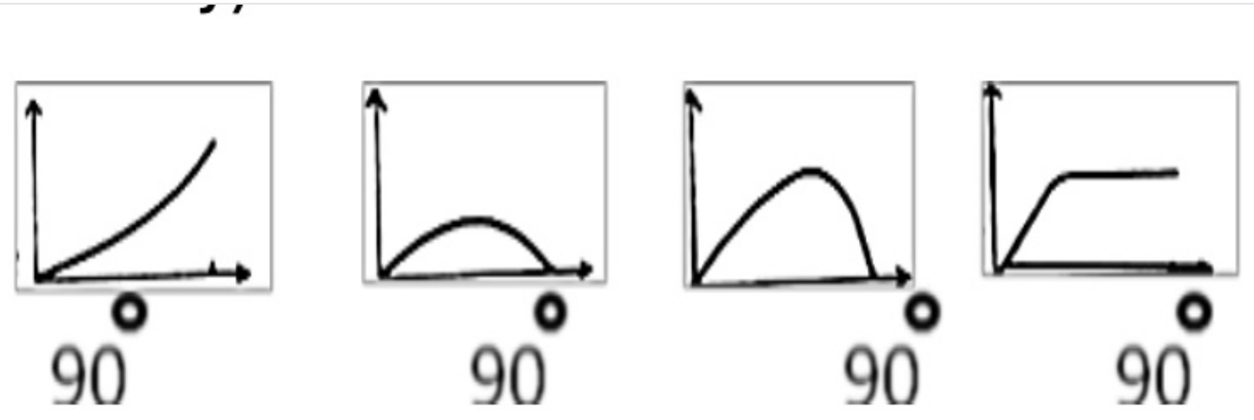
(B) Active site recognizes several molecules.

(C) Active site does not change its shape.

(D) Active site attracts & hold products.

Q175. Single correct

Which of the following graph diagrams represent the relationship between enzyme activity and temperature, if the temperature is raised to 90°C from 0°C?
(X-axis: Temperature, Y-axis: enzyme activity)



(A) A

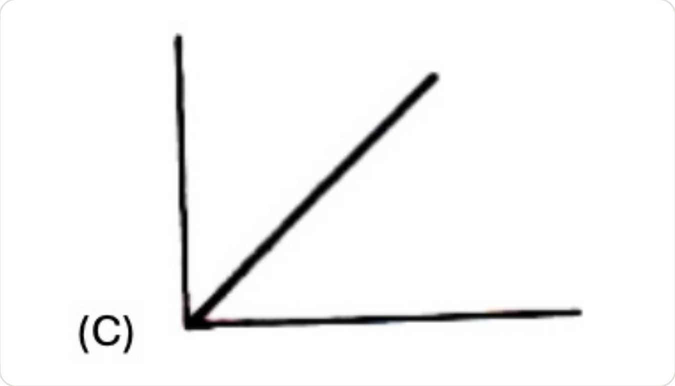
(B) B

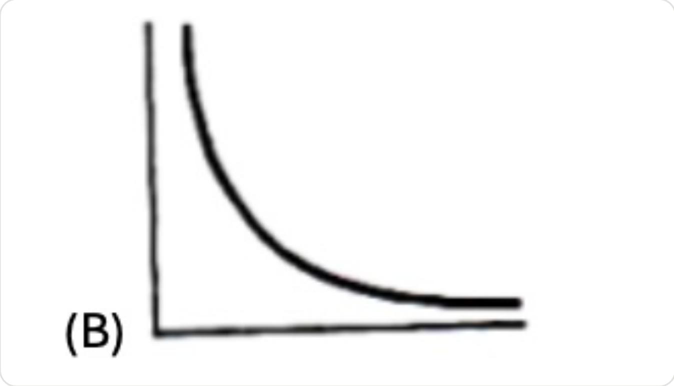
(C) C ✓

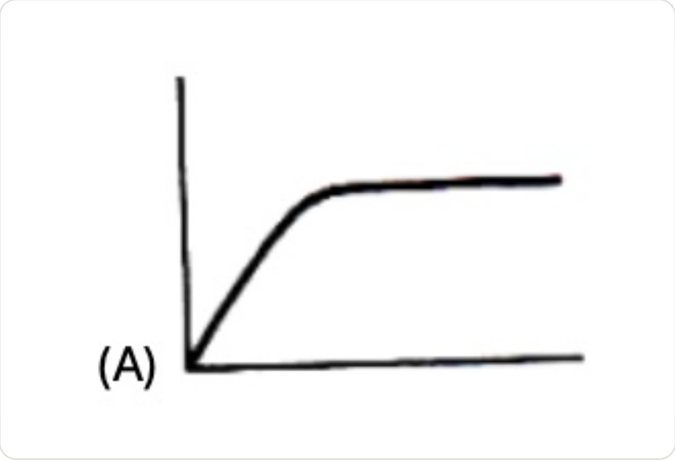
(D) D

Q176. Single correct

Which of the following graph diagrams represent the relationship between enzyme activity and substrate concentration? (X-axis: substrate concentration, Y-axis: enzyme activity)







(A) A ✓

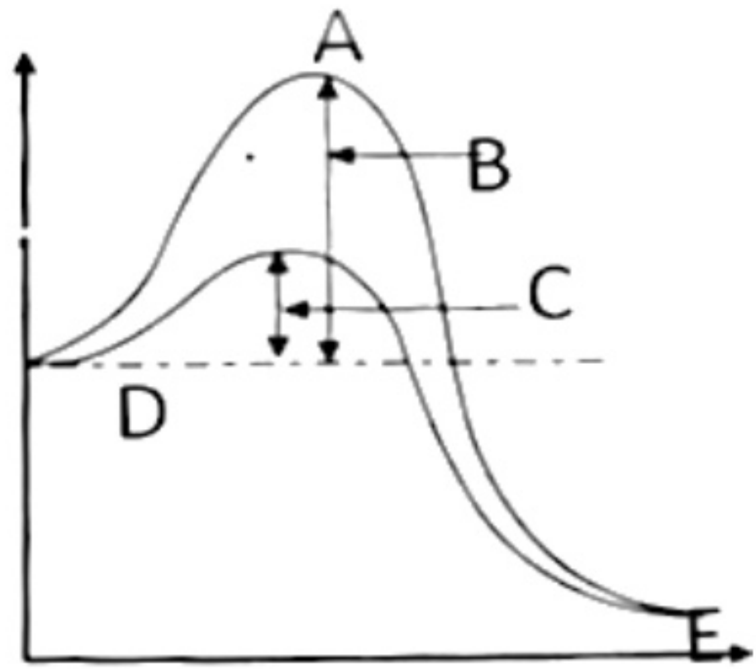
(B) B

(C) C

(D) D

Q177. Single correct

The graph below is a representation of Potential energy (Y axis) Vs Progression of structural transformation (X axis). What are the A-E in the graph?



(A) A-transition state, B-activation energy without the enzyme, C-activation energy with the enzyme, D-substrate, E-product ✓

(B) A- transition state, B- activation energy with the enzyme, C-activation energy without the enzyme, D-product, E-substrate

(C) A- transition state, B- activation energy with the enzyme, C-activation energy without the enzyme, D-substrate, E-product

(D) A-transition state, B-activation energy without the enzyme, C-activation energy with the enzyme, D-product, E-substrate

Q178. Single correct

A: Metal ion like Zn^{2+} can be regarded as cofactor.

R: Removal of co factor does not effect enzyme activity.

In the questions from 56-60 mark the following

(A) If both the assertion and the reason are true and the reason is a correct explanation of the assertion

(B) If both the assertion and reason are true but the reason is not a correct explanation of the assertion

(C) If the assertion is true but the reason is false ✓

(D) If both the assertion and reason are false

Q179. Single correct

A: Vmax refers to maximum velocity of enzyme action.

R: Vmax/2 is also knows as KM.

(A) If both the assertion and the reason are true and the reason is a correct explanation of the assertion

(B) If both the assertion and reason are true but the reason is not a correct explanation of the assertion

(C) If the assertion is true but the reason is false ✓

(D) If both the assertion and reason are false

Q180. Single correct

A: A competitive inhibitor competes with enzyme to stick with substrate.

R: Succinate is a competitive inhibitor of Pyruvate.

(A) If both the assertion and the reason are true and the reason is a correct explanation of the assertion

(B) If both the assertion and reason are true but the reason is not a correct explanation of the assertion

(C) If the assertion is true but the reason is false

(D) If both the assertion and reason are false ✓